

43.6 Mixed-model and Bayesian representations

The criteria of [Section 43.5](#) treat λ as a tuning constant to be optimized. There is another way to think about it. A quadratic penalty is, up to a constant, the logarithm of a normal density, so penalizing $\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma}$ is the same as believing in advance that $\boldsymbol{\gamma}$ is a draw from a normal distribution with precision proportional to $\mathbf{P}$. Then λ is a ratio of two variances, and a variance is not a tuning constant but a parameter the data can estimate. The machinery of [Chapter 32](#) becomes available, with a second dividend: an interval whose width knows about the bias.

43.6.1 Splitting the penalty

The obstacle is that $\mathbf{P}$ is singular. A normal prior with precision matrix $\mathbf{P}/\tau^2$ constrains nothing on the k -dimensional null space $\mathcal{N}(\mathbf{P})$ — the straight lines, for a second-order penalty — so it is not proper. The fix is to split $\boldsymbol{\gamma}$ into the part the penalty does not see, a *fixed* effect, and the rest, a *random* effect with a proper distribution.

Theorem 43.6.1 — A penalized spline is a linear mixed model

Let $\mathbf{P}$ be nonnegative definite of order m and rank $r = m - k$, write $\mathbf{P} = \mathbf{L}\mathbf{L}^\top$ with $\mathbf{L}$ of size $m \times r$ and full column rank, let $\mathbf{X}_\gamma$ be $m \times k$ with columns spanning $\mathcal{N}(\mathbf{P})$, and set $\mathbf{Z}_\gamma = \mathbf{L}(\mathbf{L}^\top \mathbf{L})^{-1}$.

- (a) (*The transformation.*) $[\mathbf{X}_\gamma, \mathbf{Z}_\gamma]$ is nonsingular, so every $\boldsymbol{\gamma}$ is uniquely $\boldsymbol{\gamma} = \mathbf{X}_\gamma \boldsymbol{\beta} + \mathbf{Z}_\gamma \mathbf{u}$; moreover $\mathbf{u} = \mathbf{L}^\top \boldsymbol{\gamma}$ and $\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma} = \mathbf{u}^\top \mathbf{u}$.
- (b) (*The mixed model.*) With $\mathbf{X} = \mathbf{B}\mathbf{X}_\gamma$ and $\mathbf{Z} = \mathbf{B}\mathbf{Z}_\gamma$, the P-spline criterion Eq. 43.3.1 is

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2 + \lambda \mathbf{u}^\top \mathbf{u}, \quad (43.6.1)$$

which is the penalized criterion Eq. 32.3.6 of the linear mixed model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}$, $\mathbf{u} \sim N_r(\mathbf{0}, \sigma_u^2 \mathbf{I})$, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ independent, with

$$\lambda = \sigma^2 / \sigma_u^2. \quad (43.6.2)$$

Its solution is Henderson's (Theorem 32.3.2): $\hat{\boldsymbol{\beta}}$ is the generalized least squares estimate and $\hat{\mathbf{u}}$ the BLUP (Theorem 32.3.1).

- (c) (*The Bayesian reading.*) Suppose σ^2 is known and $\boldsymbol{\gamma}$ has the prior with density proportional to $\exp\{-\boldsymbol{\gamma}^\top \mathbf{P} \boldsymbol{\gamma} / (2\tau^2)\}$, $\tau^2 = \sigma^2 / \lambda$ — equivalently $\boldsymbol{\beta}$ flat and $\mathbf{u} \sim N_r(\mathbf{0}, \tau^2 \mathbf{I})$. Then

$$\boldsymbol{\gamma} \mid \mathbf{Y} = \mathbf{y} \sim N_m\{\hat{\boldsymbol{\gamma}}, \sigma^2(\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P})^{-1}\}, \quad (43.6.3)$$

with $\hat{\boldsymbol{\gamma}}$ the penalized estimate of Proposition 43.3.2(a).

- (d) (*REML.*) The marginal distribution is $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{V}_\lambda)$ with $\mathbf{V}_\lambda = \mathbf{I} + \lambda^{-1} \mathbf{Z}\mathbf{Z}^\top$. Profiling σ^2 out of the restricted likelihood (Definition 32.4.1, Theorem 32.4.2) leaves

$$-2\ell_R(\lambda) = (n - k) \log \hat{\sigma}_\lambda^2 + \log|\mathbf{V}_\lambda| + \log|\mathbf{X}^\top \mathbf{V}_\lambda^{-1} \mathbf{X}| + c, \quad (43.6.4)$$

where $\hat{\sigma}_\lambda^2 = \mathbf{y}^\top \mathbf{A}_\lambda \mathbf{y} / (n - k)$ with $\mathbf{A}_\lambda = \mathbf{V}_\lambda^{-1} - \mathbf{V}_\lambda^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{V}_\lambda^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}_\lambda^{-1}$, and c does not depend on λ . Minimizing Eq. 43.6.4 over $\lambda > 0$ estimates the smoothing parameter.

Proof.

- (a) From $\mathbf{P} = \mathbf{L}\mathbf{L}^\top$ we get $\mathbf{P}\boldsymbol{\gamma} = \mathbf{0}$ iff $\|\mathbf{L}^\top \boldsymbol{\gamma}\|^2 = \boldsymbol{\gamma}^\top \mathbf{P}\boldsymbol{\gamma} = 0$ iff $\mathbf{L}^\top \boldsymbol{\gamma} = \mathbf{0}$, so $\mathcal{N}(\mathbf{P}) = \mathcal{N}(\mathbf{L}^\top) = \mathcal{C}(\mathbf{L})^\perp$ and $\mathcal{C}(\mathbf{Z}_\gamma) = \mathcal{C}(\mathbf{L}) = \mathcal{N}(\mathbf{P})^\perp$. Hence $\mathcal{C}(\mathbf{X}_\gamma)$ and $\mathcal{C}(\mathbf{Z}_\gamma)$ are orthogonal subspaces of dimensions k and r that together span $\mathbb{R}^m$, so $[\mathbf{X}_\gamma, \mathbf{Z}_\gamma]$ has rank m and the decomposition is unique. Multiplying $\boldsymbol{\gamma} = \mathbf{X}_\gamma \boldsymbol{\beta} + \mathbf{Z}_\gamma \mathbf{u}$ by $\mathbf{L}^\top$ gives $\mathbf{L}^\top \boldsymbol{\gamma} = \mathbf{0} + \mathbf{L}^\top \mathbf{L} (\mathbf{L}^\top \mathbf{L})^{-1} \mathbf{u} = \mathbf{u}$, and then $\boldsymbol{\gamma}^\top \mathbf{P}\boldsymbol{\gamma} = \|\mathbf{L}^\top \boldsymbol{\gamma}\|^2 = \mathbf{u}^\top \mathbf{u}$.
- (b) Substituting $\mathbf{B}\boldsymbol{\gamma} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$ and (a) into Eq. 43.3.1 gives Eq. 43.6.1. In the mixed model with $\mathbf{G} = \sigma_u^2 \mathbf{I}$ and $\mathbf{R} = \sigma^2 \mathbf{I}$, Henderson's criterion is $\sigma^{-2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2 + \sigma_u^{-2} \mathbf{u}^\top \mathbf{u}$, which is σ^{-2} times Eq. 43.6.1 when $\lambda = \sigma^2 / \sigma_u^2$; the two have the same minimizers.
- (c) The prior is flat in $\boldsymbol{\beta}$ and normal in $\mathbf{u}$, so the joint density of $(\mathbf{y}, \boldsymbol{\gamma})$ is proportional to $\exp[-\{\|\mathbf{y} - \mathbf{B}\boldsymbol{\gamma}\|^2 + \lambda \boldsymbol{\gamma}^\top \mathbf{P}\boldsymbol{\gamma}\} / (2\sigma^2)]$. By the algebra in the proof of Proposition 43.3.2(a), the exponent is $-\{(\boldsymbol{\gamma} - \hat{\boldsymbol{\gamma}})^\top (\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P})(\boldsymbol{\gamma} - \hat{\boldsymbol{\gamma}}) + C\} / (2\sigma^2)$ with C free of $\boldsymbol{\gamma}$. As a function of $\boldsymbol{\gamma}$ this is the normal density Eq. 43.6.3 (Definition 39.4.1).
- (d) $\text{Cov}(\mathbf{Y}) = \sigma_u^2 \mathbf{Z}\mathbf{Z}^\top + \sigma^2 \mathbf{I} = \sigma^2 \mathbf{V}_\lambda$ by Eq. 32.2.2, and Theorem 32.4.2 gives the restricted log likelihood $-\frac{1}{2} \{(n - k) \log \sigma^2 + \log|\mathbf{V}_\lambda| + \log|\mathbf{X}^\top \mathbf{V}_\lambda^{-1} \mathbf{X}| + \mathbf{y}^\top \mathbf{A}_\lambda \mathbf{y} / \sigma^2\}$ up to a constant. Differentiating in σ^2 gives $\hat{\sigma}_\lambda^2 = \mathbf{y}^\top \mathbf{A}_\lambda \mathbf{y} / (n - k)$, and substituting it back gives Eq. 43.6.4.

■

Idea. The same algebra has now been read three ways. A quadratic penalty is a shrinkage rule (Theorem 27.2.2), a normal prior (Proposition 39.5.1) and a variance component. Only the third makes the tuning constant estimable, because only the third supplies a probability model for the coefficients and with it a likelihood in λ . Nothing is obtained for nothing: that model says the true f is a draw from a particular distribution of curves, a stronger assumption than “ f is smooth”.

Remark (What the transformation looks like). For a second-order difference penalty, $\mathcal{N}(\mathbf{D}_2)$ is spanned by $\mathbf{1}$ and $(1, \dots, m)^\top$, so the two fixed-effect columns of $\mathbf{X} = \mathbf{B}\mathbf{X}_\gamma$ span the constant and the linear function (Proposition 43.3.3(c)): the mixed model has a straight line as its fixed part and the deviations from it as its random part — the decomposition a data analyst would choose by hand, forced by the penalty rather than assumed. Other choices of $\mathbf{L}$ and $\mathbf{X}_\gamma$ give the same fitted values, since they differ by a nonsingular reparameterization (Theorem 6.7.1).

43.6.2 Bands that know about the bias

Proposition 43.5.1(e) left a problem: the obvious interval is centred at $E\{\hat{f}(x_i)\}$, which is not $f(x_i)$. The posterior Eq. 43.6.3 offers a different interval, and it is wider.

Proposition 43.6.2 — Wahba–Nychka bands

Let $\mathbf{S}_\lambda = \mathbf{B}(\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P})^{-1} \mathbf{B}^\top$ with diagonal entries s_{ii} .

- (a) Under [Theorem 43.6.1\(c\)](#), $f(x_i) = \mathbf{B}_i \boldsymbol{\gamma}$ has posterior distribution $N(\hat{f}(x_i), \sigma^2 s_{ii})$, giving the **Bayesian band**

$$\hat{f}(x_i) \pm z_{1-\alpha/2} \hat{\sigma} \sqrt{s_{ii}}. \quad (43.6.5)$$

- (b) $s_{ii} \geq (\mathbf{S}_\lambda \mathbf{S}_\lambda^\top)_{ii}$, so [Eq. 43.6.5](#) is never narrower than the interval of Proposition 43.5.1(e), and is strictly wider unless $\mathbf{S}_\lambda$ is a projection.
- (c) (*Stated, not proved.*) Treating f as fixed, the extra width in (b) accounts approximately for the squared bias, in the sense that $n^{-1} \sum_i \sigma^2 s_{ii}$ approximates $n^{-1} \sum_i [\text{Var}\{\hat{f}(x_i)\} + \text{bias}\{\hat{f}(x_i)\}^2]$ when λ is near its risk-optimal value. Consequently the *average* over the design points of the coverage of [Eq. 43.6.5](#) is close to $1 - \alpha$, while the coverage at any single x_i is not: it is below nominal where f curves sharply and above it where f is nearly straight. This is **across-the-function** coverage (Wahba, 1983; Nychka, 1988).

Proof.

- (a) By [Eq. 43.6.3](#), $\mathbf{B}_i \boldsymbol{\gamma}$ is normal with mean $\mathbf{B}_i \hat{\boldsymbol{\gamma}} = \hat{f}(x_i)$ and variance $\sigma^2 \mathbf{B}_i (\mathbf{B}^\top \mathbf{B} + \lambda \mathbf{P})^{-1} \mathbf{B}_i^\top = \sigma^2 s_{ii}$.
- (b) $\mathbf{S}_\lambda$ is symmetric with eigenvalues $s_j \in (0, 1]$ by Proposition 43.3.2(b), so $\mathbf{S}_\lambda - \mathbf{S}_\lambda^2$ is symmetric with eigenvalues $s_j(1 - s_j) \geq 0$ and hence nonnegative definite; its i th diagonal entry $s_{ii} - (\mathbf{S}_\lambda \mathbf{S}_\lambda^\top)_{ii}$ is therefore nonnegative, and zero for all i only if the matrix is zero, that is only if every $s_j \in \{0, 1\}$.

Part (c) is not proved here; see the notes. ■

Warning. “Across-the-function” is a weaker guarantee than it sounds. Count how often the band contains the true curve across the design points and the fraction is about $1 - \alpha$. It does not say that the band contains the curve *everywhere* with probability $1 - \alpha$ — that is a simultaneous band, which is wider (Sun and Loader, 1994) — nor that the interval at any one chosen point has nominal coverage. If a particular x is the question, [Eq. 43.6.5](#) is not the answer.

Example. (Measuring the coverage)

Take $n = 120$ equally spaced points, the mean function of Example (Bias, variance and the optimal bandwidth) and $\sigma = 0.25$, with a cubic P-spline on $K = 20$ knots, so $m = 24$ coefficients split into $k = 2$ fixed and $r = 22$ random ones. On one data set the transformation of [Theorem 43.6.1](#) reproduces the penalized fit to machine precision, REML chooses $\lambda = 5.01 \times 10^0$, which is 8.12 effective degrees of freedom, against 7.10 for GCV; the REML estimate of σ is 0.254.

Over 400 replicates, with λ re-estimated by REML each time, the nominal 95 per cent band [Eq. 43.6.5](#) covers $f(x_i)$ at a fraction 0.954 of the design points on average, the

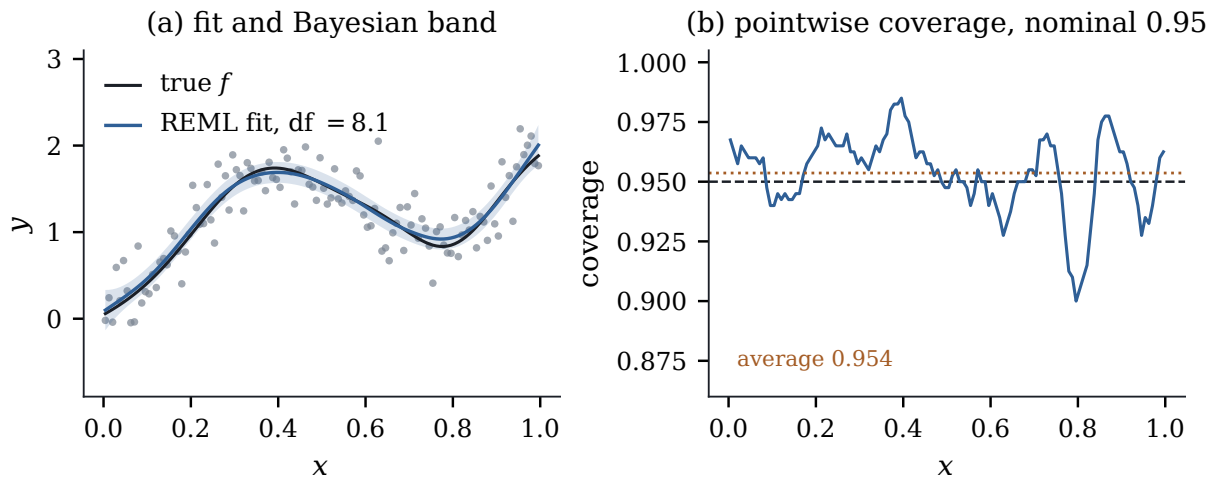


Figure 43.6.1: (a) One data set, the REML fit, and the nominal 95 per cent Bayesian band Eq. 43.6.5. (b) Coverage of that band at each design point over 400 replicates, λ re-estimated each time; dashed line nominal, dotted line the average over the design points.

pointwise coverages running from 0.900 at the narrow dip near $x = 0.8$, where the fit is most biased, to 0.985 on the flat stretches (Figure 43.6.1). The Bayesian standard errors are on average 1.12 times the naive ones of Proposition 43.5.1(e), whose bands cover only 0.927 of the design points: a twelve per cent widening buys back nearly three percentage points of average coverage, and what is left over is the bias that no single width can equalize.

```
import numpy as np

def f_true(x):
    """The mean function of Section 43.1."""
    return 2 * x + np.exp(-25 * (x - 0.35) ** 2) - 0.75 * np.exp(-50 * (x - 0.80) ** 2)

def uniform_knots(a, b, K, d):
    """K interior knots equally spaced in (a, b), continued d further steps at each end."""
    delta = (b - a) / (K + 1)
    return a + delta * np.arange(-d, K + 2 + d)

def bspline_basis(x, kn, d):
    """The B-splines of degree d on the knot vector kn, by the de Boor recurrence.
    """
    x = np.atleast_1d(np.asarray(x, float))
    B = np.array([(x >= kn[j]) & (x < kn[j + 1]) for j in range(len(kn) - 1)], float).T
    last = np.max(np.nonzero(kn[:-1] < kn[1:])[0])
    B[x == kn[-1], last] = 1.0
    for deg in range(1, d + 1):
        C = np.zeros((len(x), B.shape[1] - 1))
        for j in range(C.shape[1]):
            if kn[j + deg] > kn[j]:
                C[:, j] += (x - kn[j]) / (kn[j + deg] - kn[j]) * B[:, j]
```

```

        if kn[j + deg + 1] > kn[j + 1]:
            C[:, j] += (kn[j + deg + 1] - x) / (kn[j + deg + 1] - kn[j + 1]) * B
   [:, j + 1]
    B = C
    return B

def split_penalty(P, tol=1e-9):
    """Bases  $X_g$  of  $N(P)$  and  $Z_g$  of its complement with  $Z_g' P Z_g = I$  and  $X_g' P = 0$ ."""
    e, U = np.linalg.eigh(P)
    pos = e > tol * e.max()
    L = U[:, pos] * np.sqrt(e[pos])          #  $P = L L'$  with  $L$  of full
    column rank
    Xg = U[:, ~pos]                          # the null space of the
    penalty
    Zg = L @ np.linalg.inv(L.T @ L)
    return Xg, Zg

n, sigma, K, d, k = 120, 0.25, 20, 3, 2
x = (np.arange(1, n + 1) - 0.5) / n
kn = uniform_knots(0.0, 1.0, K, d)
B = bspline_basis(x, kn, d)
m = B.shape[1]
Dk = np.diff(np.eye(m), n=k, axis=0)
P = Dk.T @ Dk
Xg, Zg = split_penalty(P)
X, Z = B @ Xg, B @ Zg
r = Z.shape[1]

rng = np.random.default_rng(431)
g = rng.normal(size=m)                      # any coefficient
vector
beta_u = np.linalg.solve(np.column_stack([Xg, Zg]), g)
print("the penalty, in the old and the new coordinates:", g @ P @ g, beta_u[k:] @
      beta_u[k:])

y = f_true(x) + sigma * rng.normal(size=n)
lam = 1.0
fit_pen = B @ np.linalg.solve(B.T @ B + lam * P, B.T @ y)
C = np.block([[X.T @ X, X.T @ Z], [Z.T @ X, Z.T @ Z + lam * np.eye(r)]]) #
Henderson's equations
sol = np.linalg.solve(C, np.concatenate([X.T @ y, Z.T @ y]))
fit_mix = X @ sol[:k] + Z @ sol[k:]
print("penalized fit versus mixed-model fit:", np.max(np.abs(fit_pen - fit_mix)))

```

[Run this code online](#)

```

Uz, sz, _ = np.linalg.svd(Z, full_matrices=False)    # the design is fixed, so do
this once

def reml(y, lam):
    """Profiled REML criterion (to be minimized) for  $V = I + Z Z' / \text{lam}$ ."""
    w = sz**2 / (lam + sz**2)
    def Vi(M):
        M2 = M if M.ndim == 2 else M[:, None]          # multiplication by  $V^{-1}$ 

```

```

    out = M2 - Uz @ (w[:, None] * (Uz.T @ M2))
    return out if M.ndim == 2 else out[:, 0]

logdetV = np.sum(np.log1p(sz**2 / lam))
ViX = Vi(X)
A = X.T @ ViX
resid = Vi(y) - ViX @ np.linalg.solve(A, ViX.T @ y)
s2 = y @ resid / (n - k)
return 0.5 * ((n - k) * np.log(s2) + logdetV + np.linalg.slogdet(A)[1]), s2

def reml_lambda(y, grid):
    return grid[int(np.argmin([reml(y, l)[0] for l in grid]))]

grid = np.geomspace(1e-6, 1e6, 121)
lam_reml = reml_lambda(y, grid)
S_reml = B @ np.linalg.solve(B.T @ B + lam_reml * P, B.T)
df_reml = np.trace(S_reml)
gcvs = []
for l in grid:
    S = B @ np.linalg.solve(B.T @ B + l * P, B.T)
    gcvs.append(np.mean((y - S @ y) ** 2) / (1 - np.trace(S) / n) ** 2)
lam_gcv = grid[int(np.argmin(gcvs))]
df_gcv = np.trace(B @ np.linalg.solve(B.T @ B + lam_gcv * P, B.T))
print(f"REML: lambda = {lam_reml:.3g}, df = {df_reml:.2f}; GCV: df = {df_gcv:.2f}")

```

[Run this code online](#)

43.6.3 REML against the criteria, and full Bayes

REML and GCV estimate different things. GCV aims at prediction error; REML maximizes a likelihood under the assumption that f is a draw from the prior — which nobody believes, since nobody believes the deviations from a straight line are exchangeable normal increments. In practice the two rarely differ by much: in [Example \(Measuring the coverage\)](#) REML chose 8.12 effective degrees of freedom against GCV's 7.10. But REML is the more stable, its criterion better determined near the optimum, and it almost never produces the interpolating fits that GCV occasionally does (Reiss and Ogden, 2009). It has no answer when the error model is wrong: with the AR(1) errors of [Example \(What correlated errors do\)](#) it overfits just as GCV does, unless the correlation is put into $\text{Cov}(\epsilon)$, which the mixed-model form at least makes possible (Definition 33.4.1).

Remark (Full Bayes). [Theorem 43.6.1\(c\)](#) fixes λ . A fully Bayesian treatment puts a prior on τ^2 and σ^2 too — conventionally inverse-gamma, which is conditionally conjugate — and samples $(\gamma, \tau^2, \sigma^2)$ by the Gibbs sampler of Definition 39.4.2: each sweep draws γ from [Eq. 43.6.3](#), then τ^2 from an inverse-gamma with shape $a + r/2$ and scale $b + \mathbf{u}^\top \mathbf{u}/2$, then σ^2 likewise. The posterior for f then averages over the smoothing parameter instead of conditioning on an estimate of it, the cleanest answer to the selection problem of [Section 43.5](#). It is also where the prior shows its weakness: the posterior for τ^2 is sensitive to the hyperparameters, as Proposition 39.5.1 warns. [Chapter 44](#) uses this machinery for several smooth terms at once.

43.6.4 What this book does not do with smoothing

Three threads end here rather than continue, and it is only fair to name them.

Adaptive smoothing. A single λ is a compromise when f is rough in one place and flat in another; letting it vary with x — a second smooth for $\log \lambda$, free-knot splines (Denison, Mallick and Smith, 1998), wavelet thresholding (Donoho and Johnstone, 1994) — is a subject this book does not enter.

Simultaneous bands. Eq. 43.6.5 is pointwise, or at best across-the-function; a band containing the whole curve with a stated probability needs the tube formula of Sun and Loader (1994) or a bootstrap over the whole fit.

Shape constraints. Monotone, convex or unimodal fits add linear inequality constraints to Eq. 43.3.1, a quadratic program; Ramsay (1988) treats the monotone case.

What does continue is the penalty. Everything after Section 43.2 rests on one idea — a rich basis, a quadratic penalty, a trace that counts what was spent, and a variance ratio that can be estimated — and that idea is indifferent to how many covariates there are, to whether the response is normal, and to which parameter of the response distribution is modelled. Chapter 44 adds covariates and Chapter 45 parameters, both by writing down more terms of the form Eq. 43.3.1.

43.6.5 Exercises

43.6.5.1 A. Check your understanding

Exercise A1.

Using Eq. 43.6.2, say what $\sigma_u^2 \rightarrow 0$ and $\sigma_u^2 \rightarrow \infty$ correspond to in Proposition 43.3.2, and what a REML estimate $\hat{\sigma}_u^2 = 0$ would mean about the fitted curve. Is that estimate on the boundary of the parameter space, and does Theorem 32.5.2 apply?

Solution. $\sigma_u^2 \rightarrow 0$ gives $\lambda \rightarrow \infty$, leaving only the fixed part, the straight line; $\sigma_u^2 \rightarrow \infty$ gives $\lambda \rightarrow 0$, the unpenalized spline fit. A REML estimate of zero means the data give no evidence of departure from a straight line, and the reported fit *is* the least squares line. It is on the boundary, so a likelihood ratio test of $\sigma_u^2 = 0$ has not the usual χ^2 null distribution but the mixture of Theorem 32.5.2 — the same caution as for testing whether a smooth term is needed at all.

43.6.5.2 B. Practice

Exercise B1.

Run the simulation of Example (The three criteria on one data set, and over 200 of them) again with λ chosen by REML, and compare the median selected degrees of freedom, the average squared error and the upper tail of the df distribution with the values for CV, GCV and AIC_c . Which criterion is safest, and at what cost?

Exercise B2.

[Example \(Measuring the coverage\)](#) uses $\hat{\sigma}$ from Eq. 43.5.2 in [Eq. 43.6.5](#). Repeat the coverage simulation with σ known, and with $z_{1-\alpha/2}$ replaced by a t quantile on $n - 2 \operatorname{tr}(\mathbf{S}) + \operatorname{tr}(\mathbf{S}\mathbf{S}^\top)$ degrees of freedom. How much of the departure from nominal coverage is due to estimating σ , and how much to the bias?

43.6.5.3 C. Going deeper**Exercise C1.**

Show that in the mixed model of [Theorem 43.6.1\(b\)](#), the BLUP satisfies $\hat{\mathbf{u}} = (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I})^{-1} \mathbf{Z}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$, and hence that $\hat{\mathbf{u}}$ is a ridge estimate ([Theorem 27.2.2](#)) on the residuals from the fixed part. Explain, using [Theorem 32.3.1](#), why the shrinkage factor is $\sigma_u^2 / (\sigma_u^2 + \sigma^2/d_j^2)$ in the j th singular direction of $\mathbf{Z}$, and what happens to it as n grows with r fixed.

Solution. The second block of Henderson's equations is $\mathbf{Z}^\top \mathbf{X}\hat{\boldsymbol{\beta}} + (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I})\hat{\mathbf{u}} = \mathbf{Z}^\top \mathbf{y}$, which rearranges to the displayed formula; it is [Eq. 27.2.1](#) with response $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ and ridge parameter $\lambda = \sigma^2/\sigma_u^2$. With $\mathbf{Z} = \sum_j d_j \mathbf{u}_j \mathbf{v}_j^\top$, [Theorem 27.2.2\(a\)](#) gives the factor $d_j^2/(d_j^2 + \lambda) = \sigma_u^2 / \{\sigma_u^2 + \sigma^2/d_j^2\}$, which is [Eq. 32.3.3](#): the random effect is shrunk according to how much information the data carry about it. As n grows with the basis fixed, every d_j^2 grows like n and the factors tend to one, which is why a low-rank smoother with a fixed basis is asymptotically a parametric fit unless m grows with n .

Exercise C2.

The prior of [Theorem 43.6.1\(c\)](#) is improper on $\mathcal{N}(\mathbf{P})$. Show that the posterior [Eq. 43.6.3](#) is nevertheless proper, provided $\mathbf{B}\mathbf{X}_\gamma$ has full column rank. By [Exercise \(C2\)](#) that holds for $k \leq 2$ whenever the design has at least k distinct points, however the knots fall; show that it fails when it has fewer — for $k = 2$, a design concentrated at a single x — and say what the posterior then does in the unidentified direction. Compare with the discussion of improper priors in [Section 39.5](#).